
ASSIGNMENT 1

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Question-2

1 Universal Approximation Theorem

1.1 (a) Formal Statement

The **Universal Approximation Theorem** states that a feedforward neural network with a single hidden layer and a non-polynomial activation function can approximate a wide class of functions arbitrarily well.

Let $K \subset \mathbb{R}^n$ be a compact set and let $f : K \rightarrow \mathbb{R}$ be a continuous function. Suppose $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous non-polynomial activation function (e.g., sigmoid).

Then for every $\epsilon > 0$, there exists a neural network of the form

$$F(x) = \sum_{i=1}^m \alpha_i \sigma(w_i^T x + b_i)$$

such that

$$\sup_{x \in K} |f(x) - F(x)| < \epsilon$$

where

- m is the number of hidden neurons,
- $w_i \in \mathbb{R}^n$ are weight vectors,
- $b_i \in \mathbb{R}$ are biases,
- $\alpha_i \in \mathbb{R}$ are output layer weights.

Thus, a single hidden layer neural network can approximate any continuous function on a compact domain arbitrarily well.

Key elements:

- **Target functions:** Continuous functions $f \in C(K)$
- **Domain assumption:** K must be compact (closed and bounded)
- **Approximation notion:** Uniform approximation
- **Norm used:** Supremum norm

$$\|f - F\|_{\infty} = \sup_{x \in K} |f(x) - F(x)|$$

1.2 (b) Intuition Behind the Proof

The main intuition behind the proof is that neural networks can construct flexible basis functions that combine to approximate complex functions.

Each hidden neuron computes a function of the form

$$\sigma(w^T x + b)$$

which acts as a nonlinear basis function. By taking linear combinations of many such functions, the network can approximate arbitrary shapes.

One way to understand this is through connections to classical approximation theory, particularly the **Stone–Weierstrass theorem**. This theorem states that certain families of functions can approximate any continuous function on a compact domain.

Sigmoid functions can behave like smooth step functions. By combining many such step-like components, the neural network can build piecewise approximations of any continuous function.

Thus, the hidden neurons act as building blocks, and their weighted combination allows the network to approximate complex functions arbitrarily closely.

1.3 (c) Limitations of the Universal Approximation Theorem

Although the theorem guarantees representational power, it has important practical limitations.

1. No guarantee on number of neurons

The theorem does not specify how many hidden neurons are required to achieve a desired approximation accuracy. In practice, a very large number of neurons may be required, making the network inefficient and computationally expensive.

This is important when designing neural networks because extremely wide networks can be difficult to train and may lead to overfitting.

2. No guarantee that training will find the solution

The theorem only guarantees that a good approximation exists, but it does not ensure that gradient-based optimization methods will find the corresponding weights.

Training neural networks involves non-convex optimization, which may get stuck in poor local minima or saddle points. Therefore, even if a network theoretically has enough capacity, it may still fail to learn the desired function during training.

1.4 (d) Effect of ReLU and Increasing Depth

When using ReLU activation functions, the universal approximation property still holds. A sufficiently wide ReLU network can approximate any continuous function on a compact domain.

However, modern deep learning often prefers **deeper networks with relatively small width**. Increasing depth allows networks to represent hierarchical compositions of functions.

Deep ReLU networks can approximate certain classes of functions much more efficiently than shallow networks. In many cases, a deep network can represent the same function using exponentially fewer neurons than a single hidden layer network.

Therefore, depth provides a more parameter-efficient way to approximate complex functions while keeping the network width relatively small.